MACHINE LEARNING & AI - COURSE

WEEK 1 – ML Fundamentals + Data Preprocessing

🔍 Focus: Basics of ML, Data Cleaning, Feature Engineering

🧠 Topics Covered:

- What is Machine Learning? (Types: Supervised, Unsupervised, Reinforcement)
- Importing & exploring datasets
- Handling missing data, encoding, scaling
- Train/Test Split

Assignments:

1. Load a dataset using Pandas and summarize basic stats (`.info()`, `.describe()`).
2. Handle missing data using mean/median imputation.
3. Encode categorical variables using LabelEncoder & OneHotEncoder.

Mini Project 1: “Titanic Survival Prediction – Data Cleaning Project”

Dataset:

Titanic Dataset (Kaggle)

Tasks:

- Clean missing data
- Encode `Sex`, `Embarked` columns
- Visualize age distribution (Matplotlib/Seaborn)
- Output cleaned dataset as new CSV